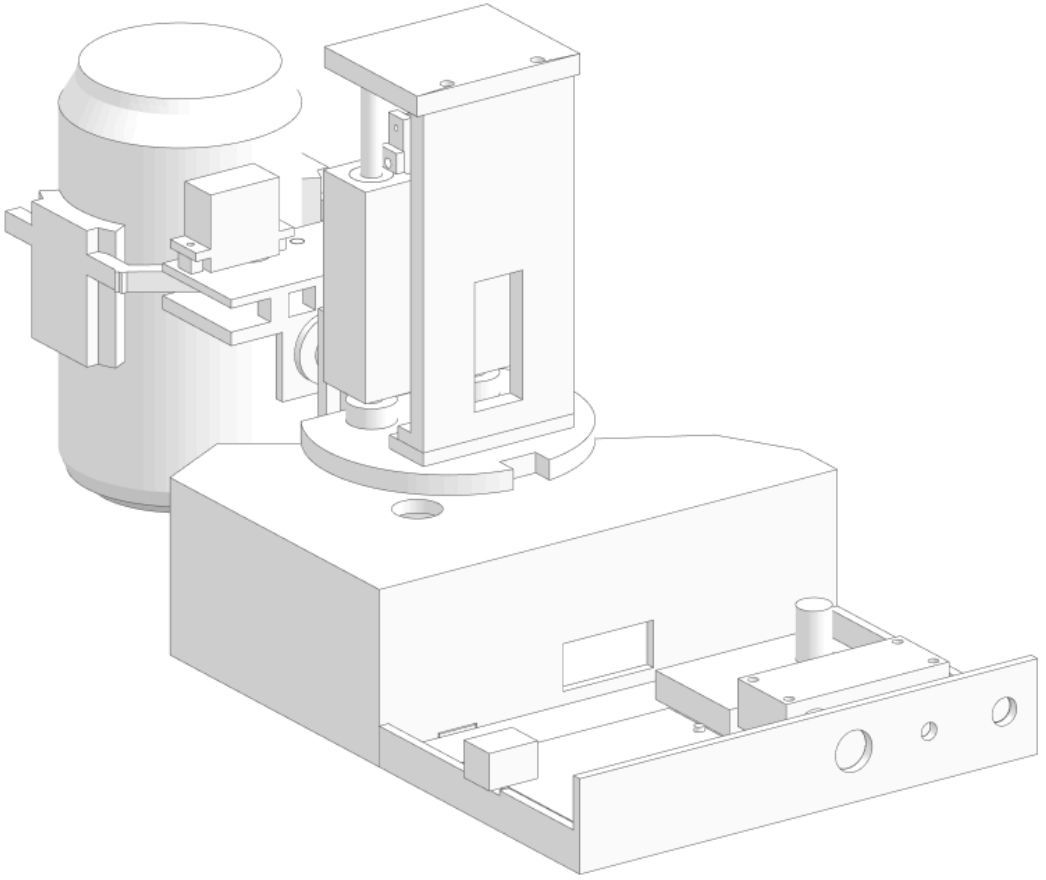
CONCEPT A

Can-flipping robot · assembly manual · MF 2331/I



1 person · ~3 h after printing · read each page, then build. **Servo angles:** load `firmware/servo_setup` and type the command in the round badge (Serial Monitor, 9600).

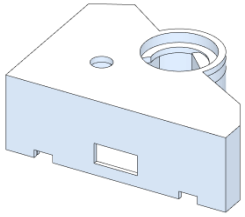
You need

Tool
Phillips screwdrivers #1 and #2
Multimeter
Hot-glue gun
Superglue
Wire stripper
Small flat screwdriver (buck pot)
Hacksaw (6 mm rod) + file
Vise or clamp (bearing press)
12 mm drill bit, turned by hand

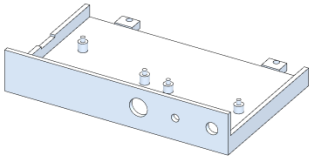
Hardware	Qty
 #6-32 × ½″	8
 #8-32 × ½″	4
 servo-bag screws	~14
6807-2RS bearing	1
LM6UU linear bearing	4
6 mm rod, cut 2 × 112 mm	2
Servos: MG996R / MG90S / SG90	1 / 2 / 1

! Before building: bench-test the electronics (docs/13_electronics_bench_test.md) and set the buck to **5.8 V**.

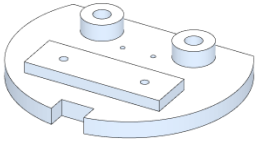
Printed parts



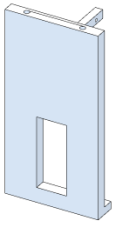
CA-101
Base



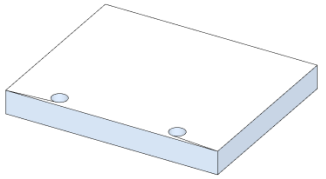
CA-102
Tray



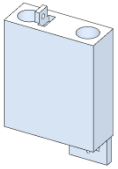
CA-201
Turret plate



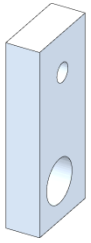
CA-202
Spine



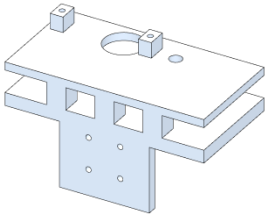
CA-203
Top cap



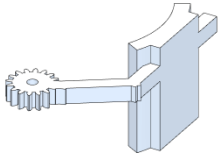
CA-301
Carriage arm



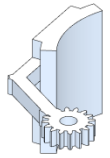
CA-302
Lift link



CA-401
Gripper frame



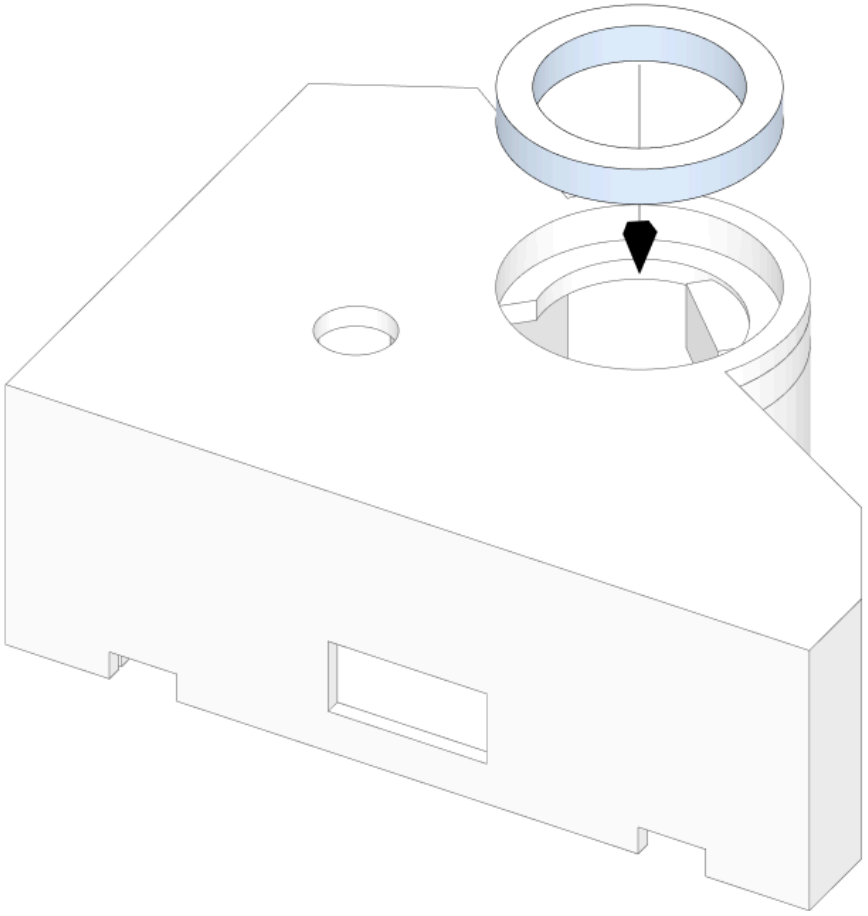
CA-402
Finger R



CA-403
Finger L

1

Bearing into the base

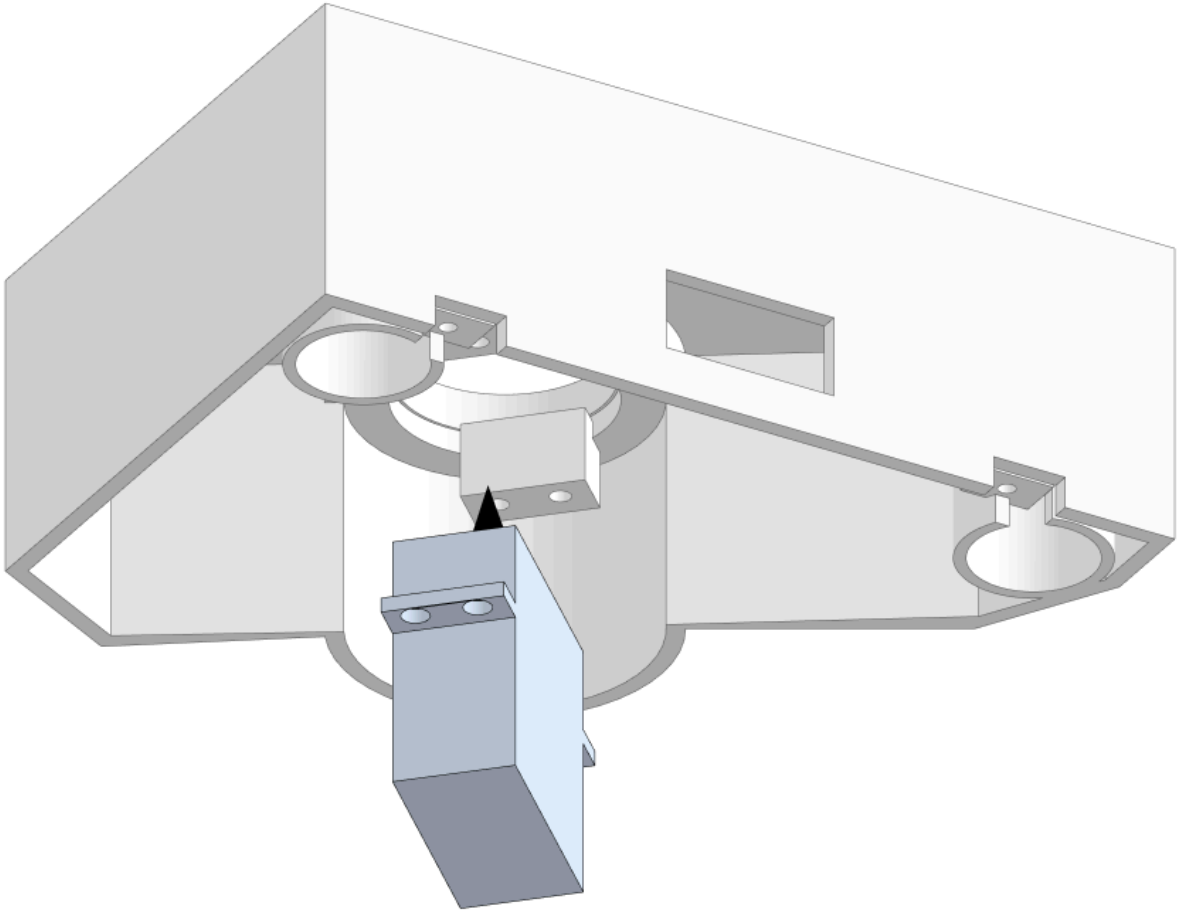


Press on the outer ring only, evenly (flat block + vise)

2

Turntable servo

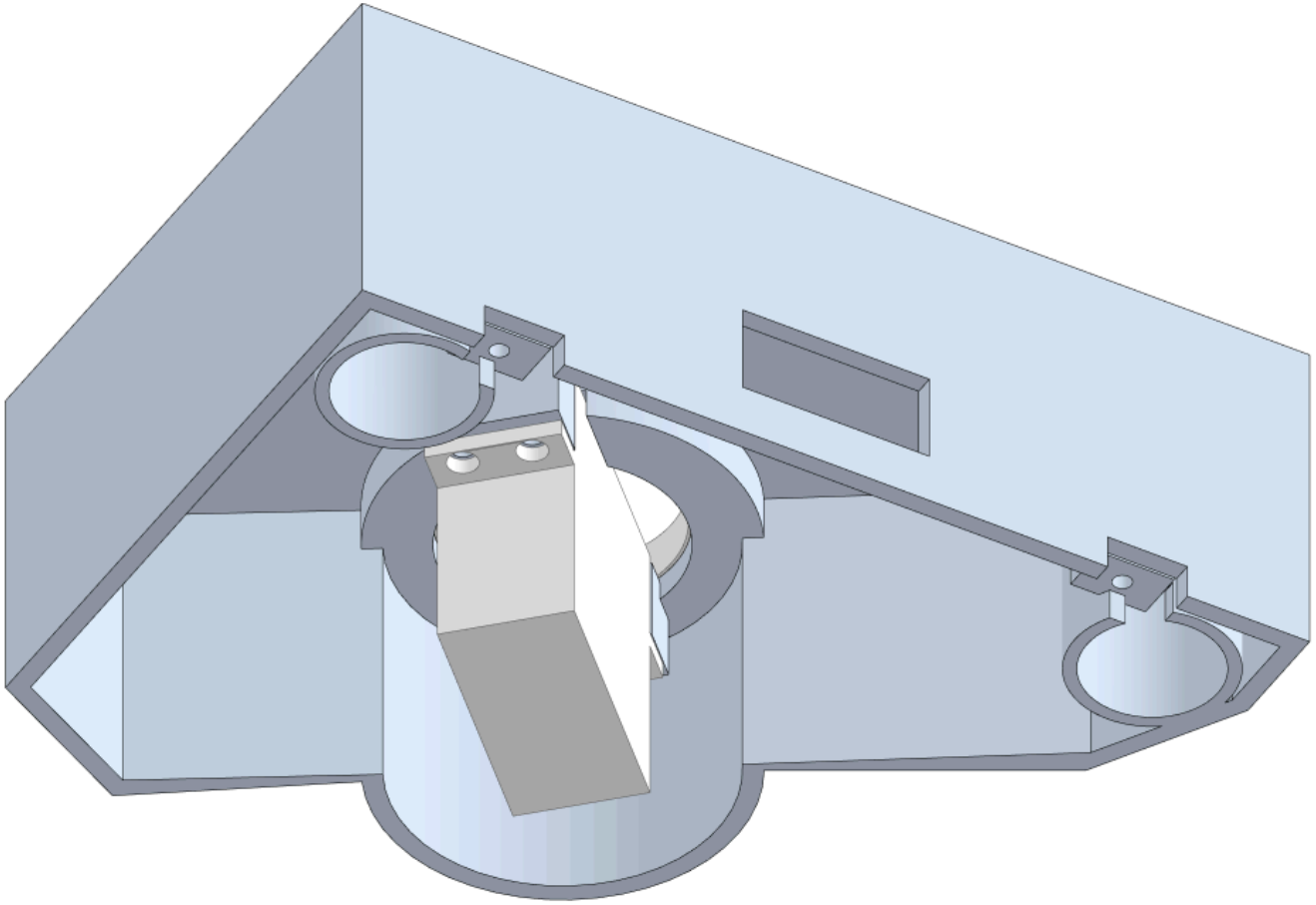
4× #8-32 × ½″



From underneath; body points away from the arm side

3

Weight + feet

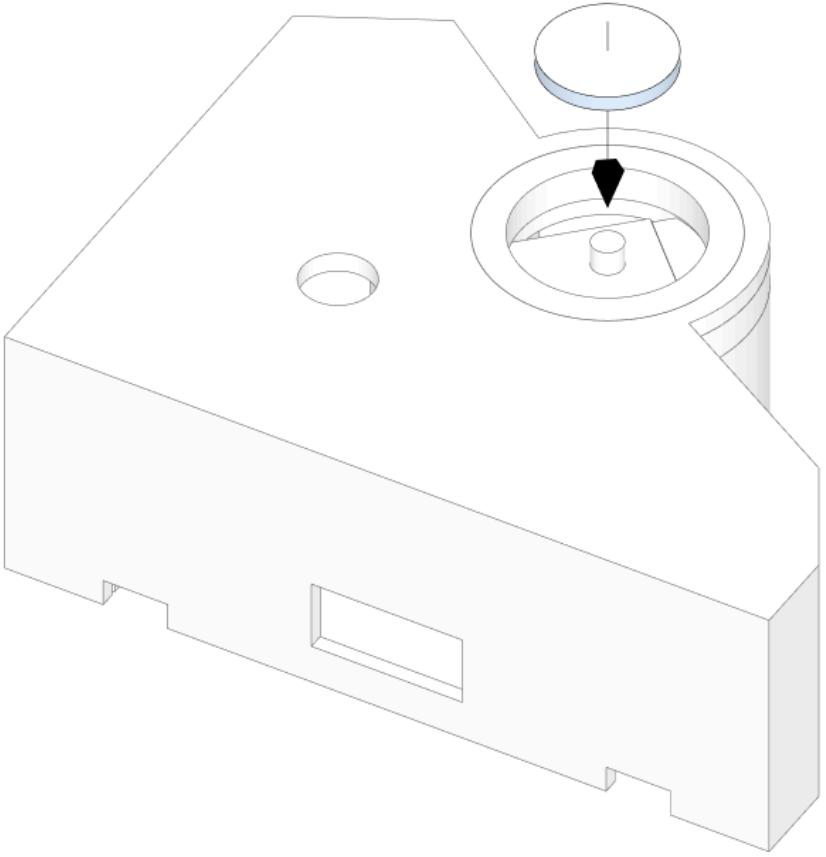


(!) ~100 g in the 2 back pockets (washers, pennies, rocks): GLUE or TAPE it in. 4 feet or hot-glue dots

4

Servo horn

1× servo-bag



MG996R

135°

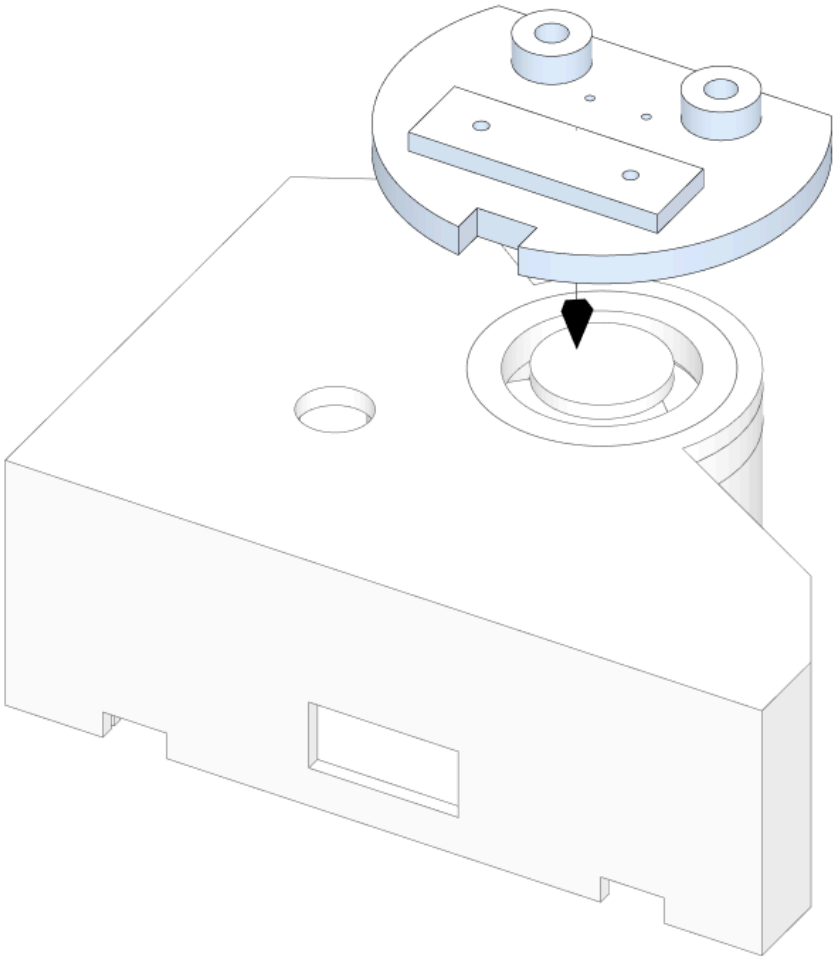
`t 135`

! Set the servo to 135° first (servo_setup: t 135)

5

Turret plate

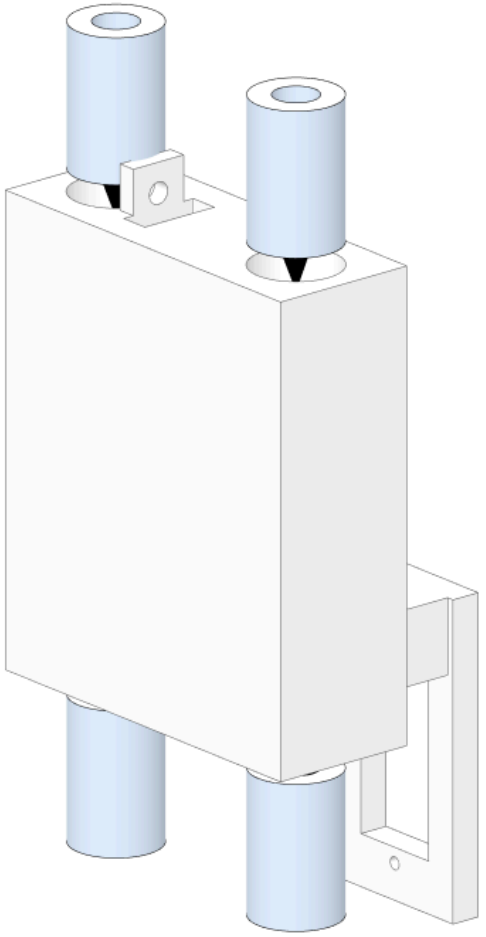
4× servo-bag



Hub into the bearing; rod bosses point at the START can. Turns 90° freely

6

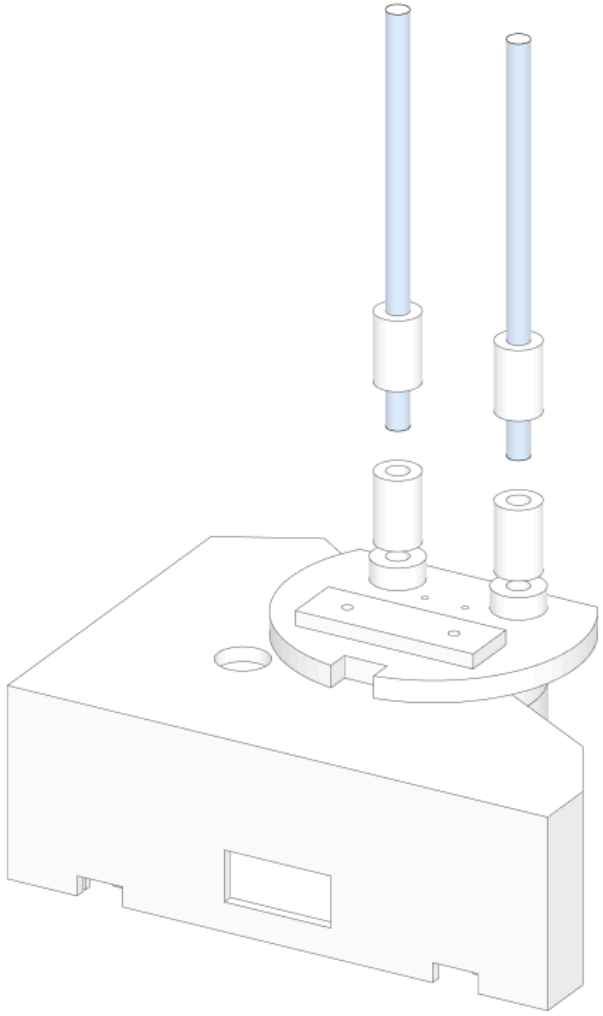
Linear bearings into the carriage



2 from below, 2 from above, flush with the faces

7

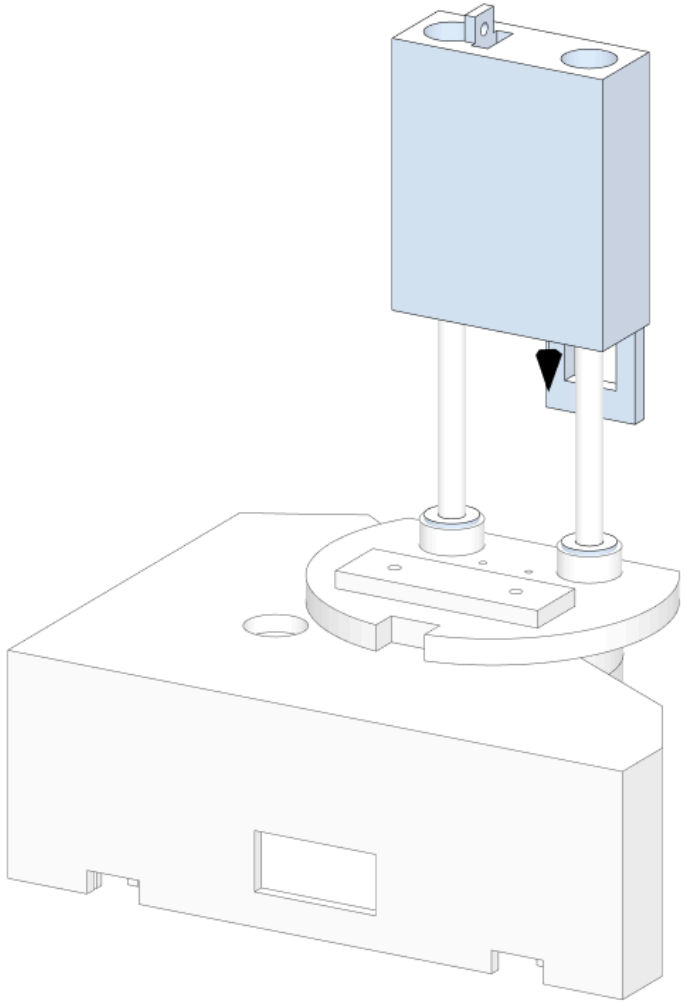
Rods



Press into the blind holes of the turret plate

8

Carriage onto the rods

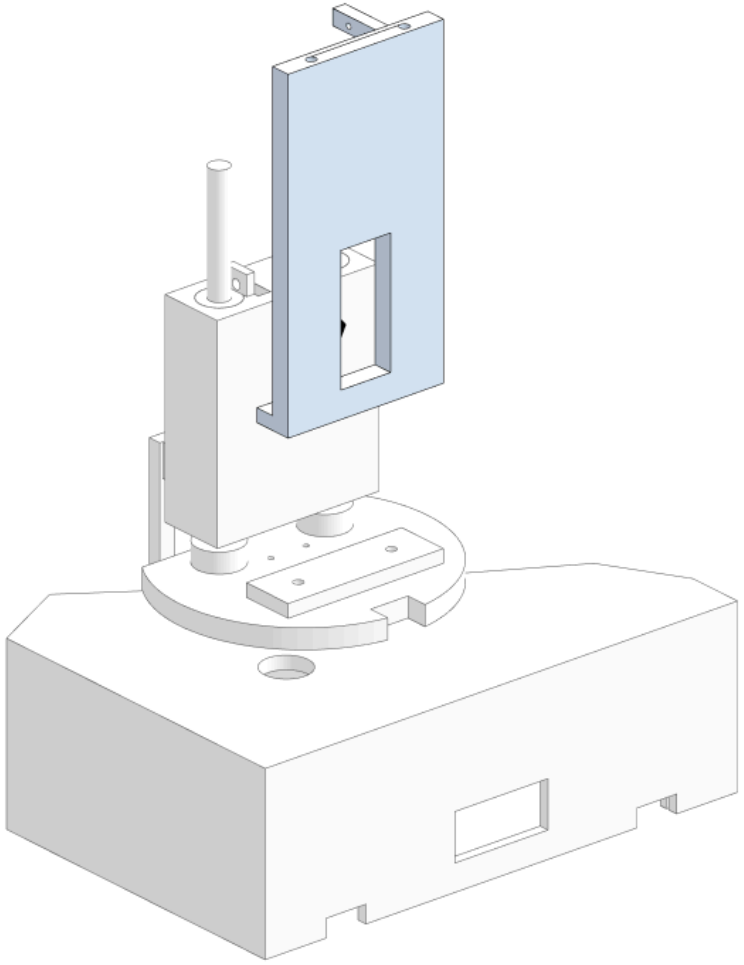


(!) It must slide down under its own weight

9

Spine

2× #6-32 × ½″

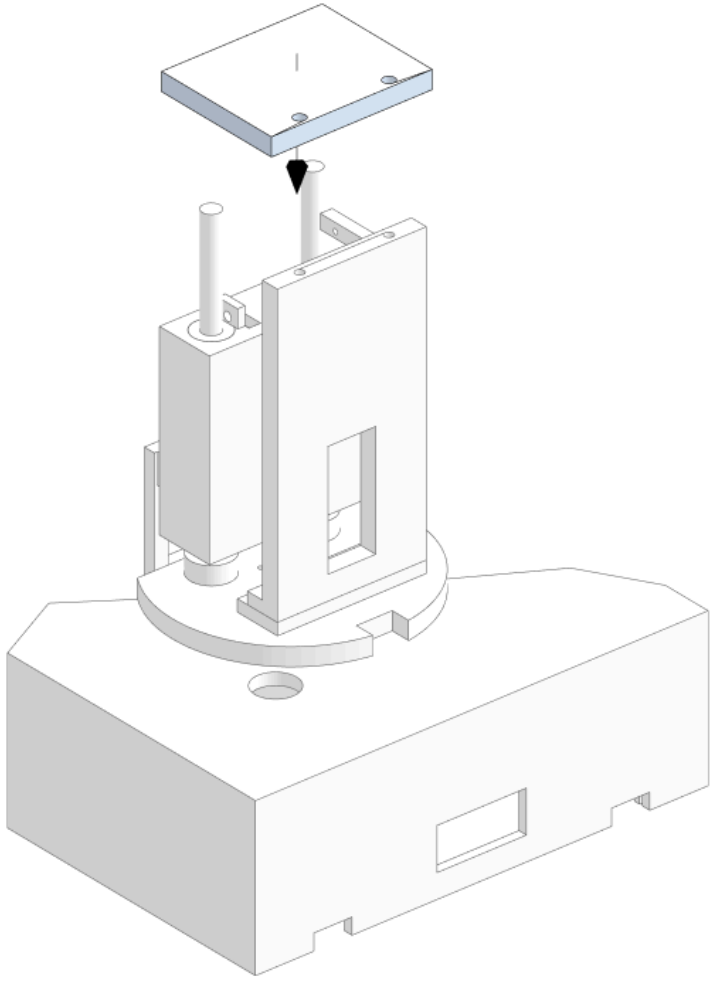


Onto the turret plate behind the rods, screws from above

10

Top cap

2× #6-32 × ½″



Over the rod tops onto the spine. Carriage still slides freely?

11

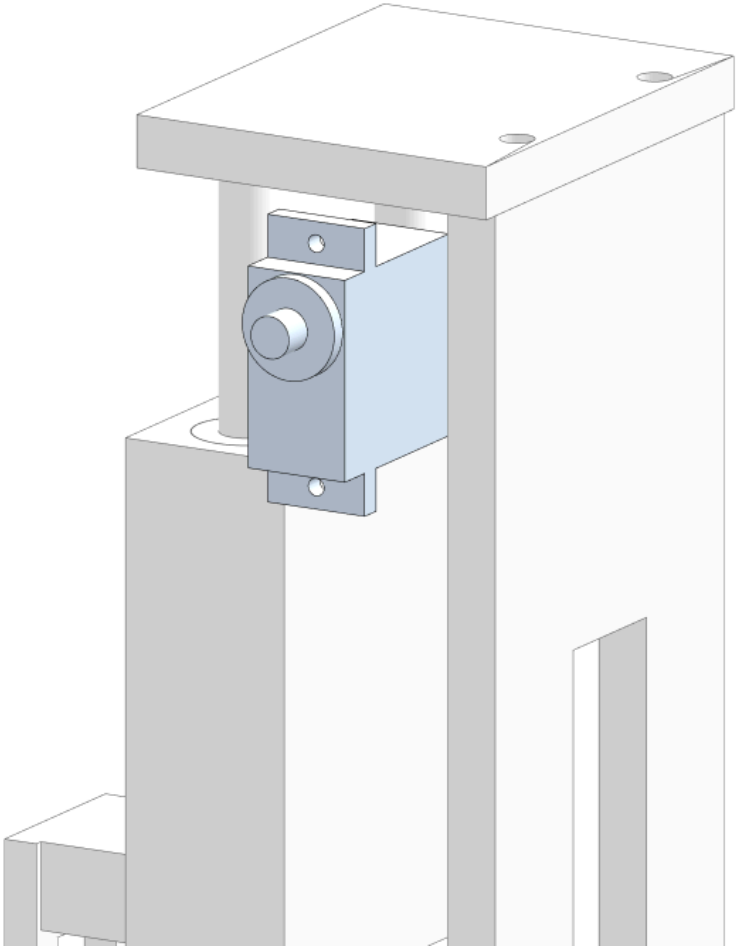
Lift servo

2× servo-bag

MG90S lift

30°

`l 30`

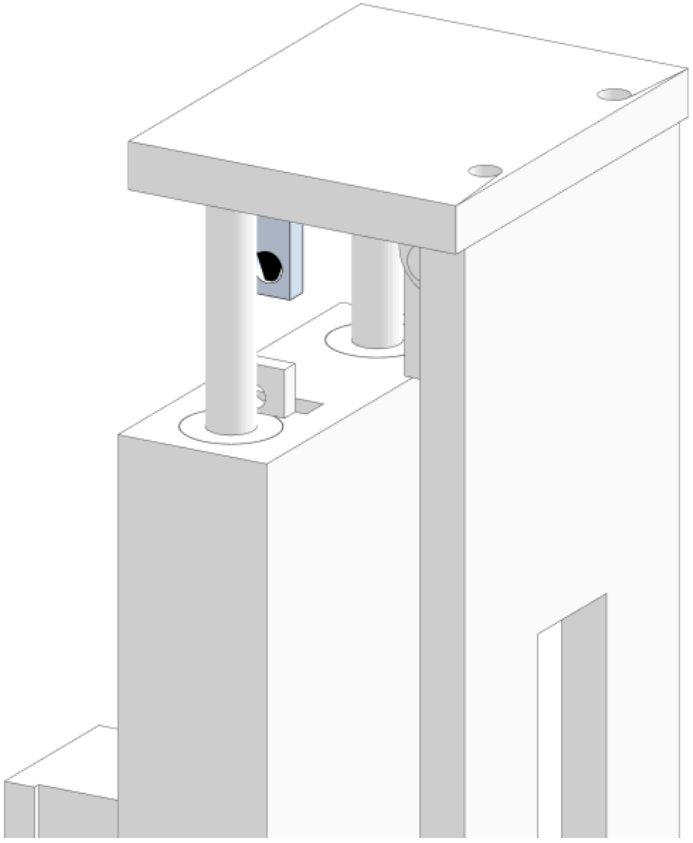


Body through the spine bracket from the left

12

Lift link

1× #6-32 × ½″



! Carriage end: #6-32 as the pivot, don't over-tighten: it must turn

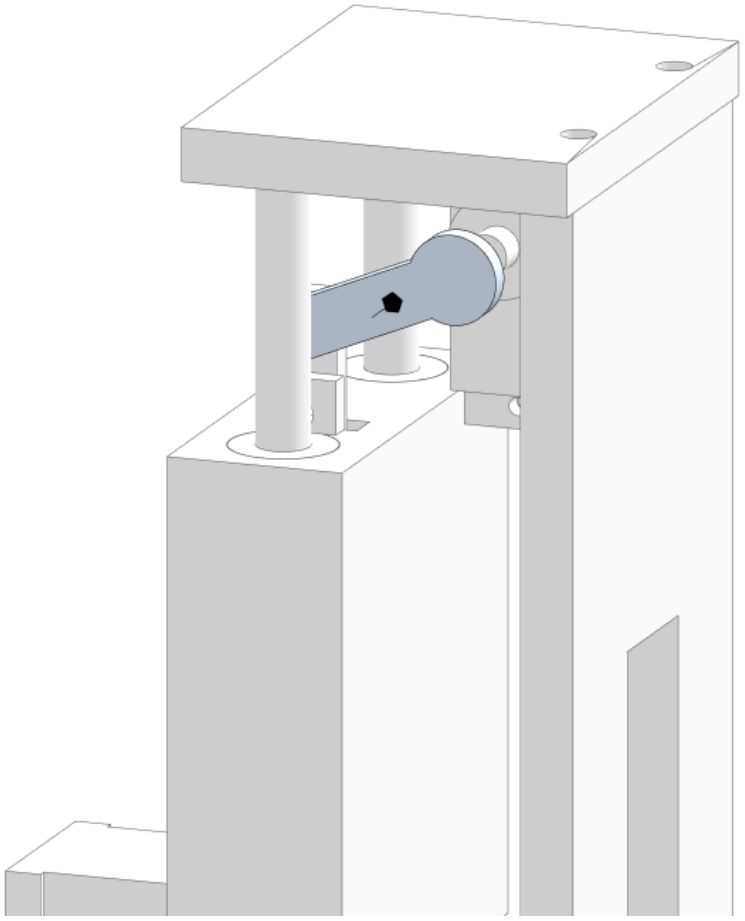
13 Crank

2× servo-bag

MG90S lift

30°

`l 30`



Single-arm horn trimmed to 20 mm, on the spline at 30°; pin screw into the link

14

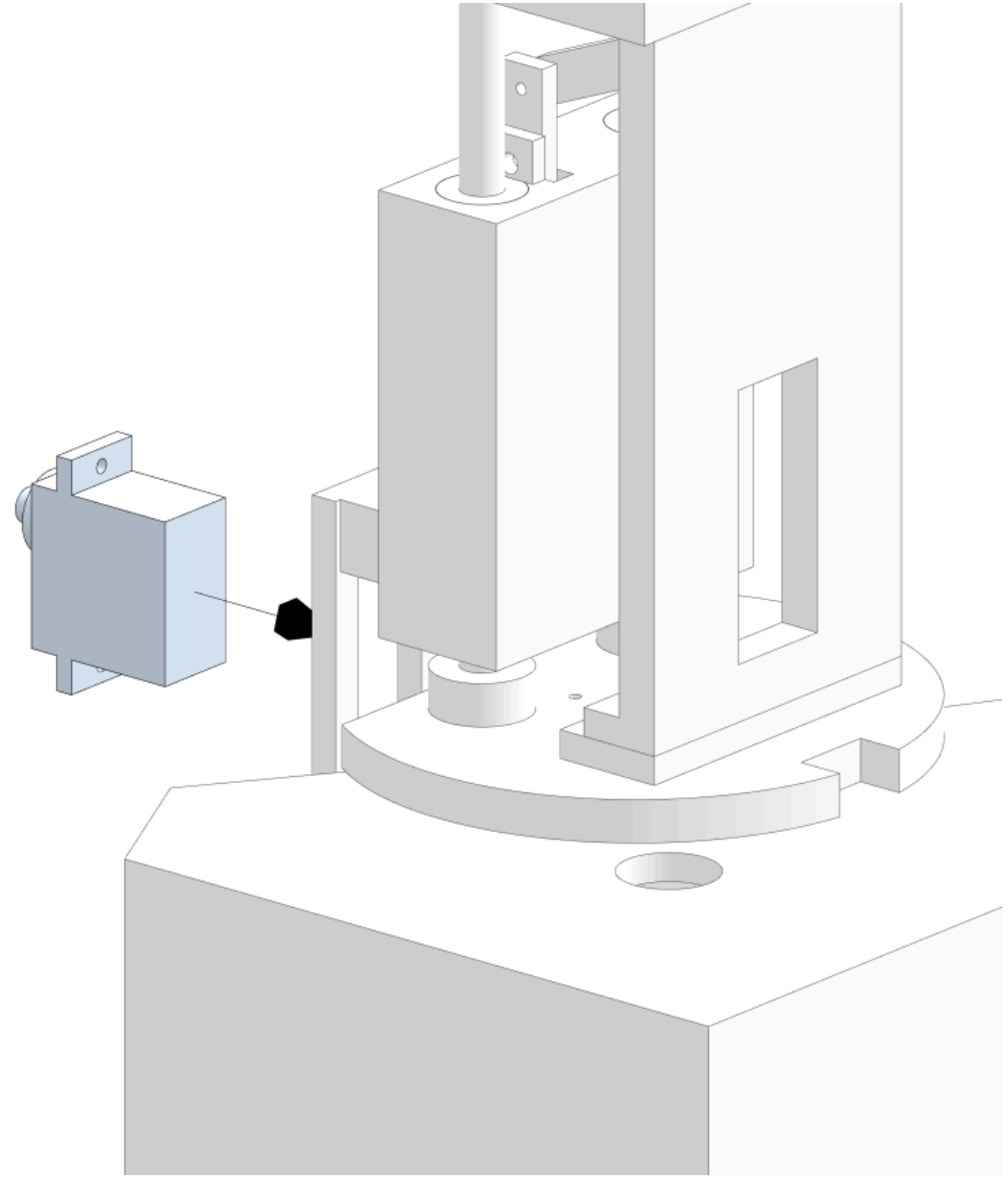
Wrist servo

2×  servo-bag

MG90S wrist

0°

`w 0`



Body through the bracket from the front

15

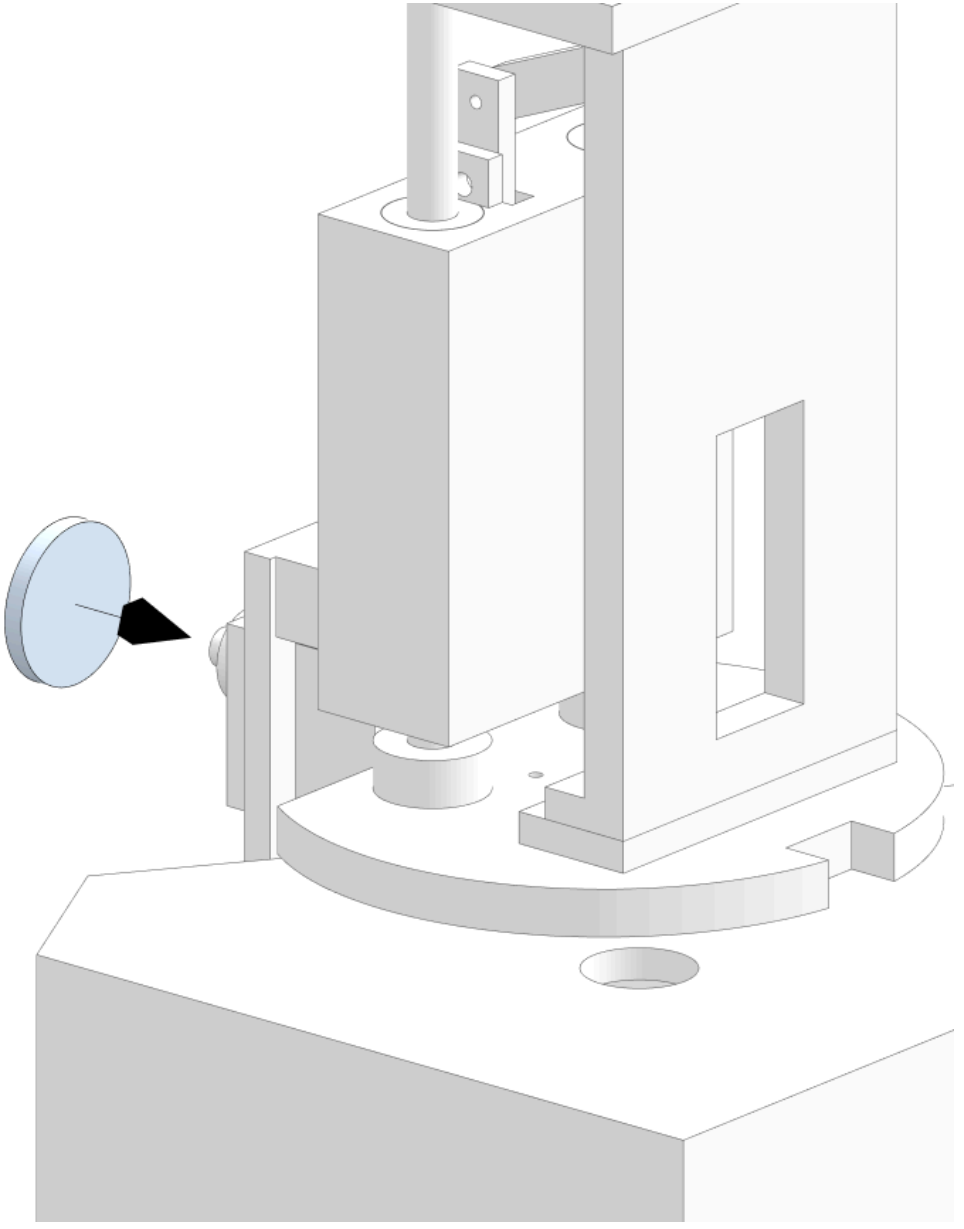
Wrist horn

1×  servo-bag

MG90S wrist

0°

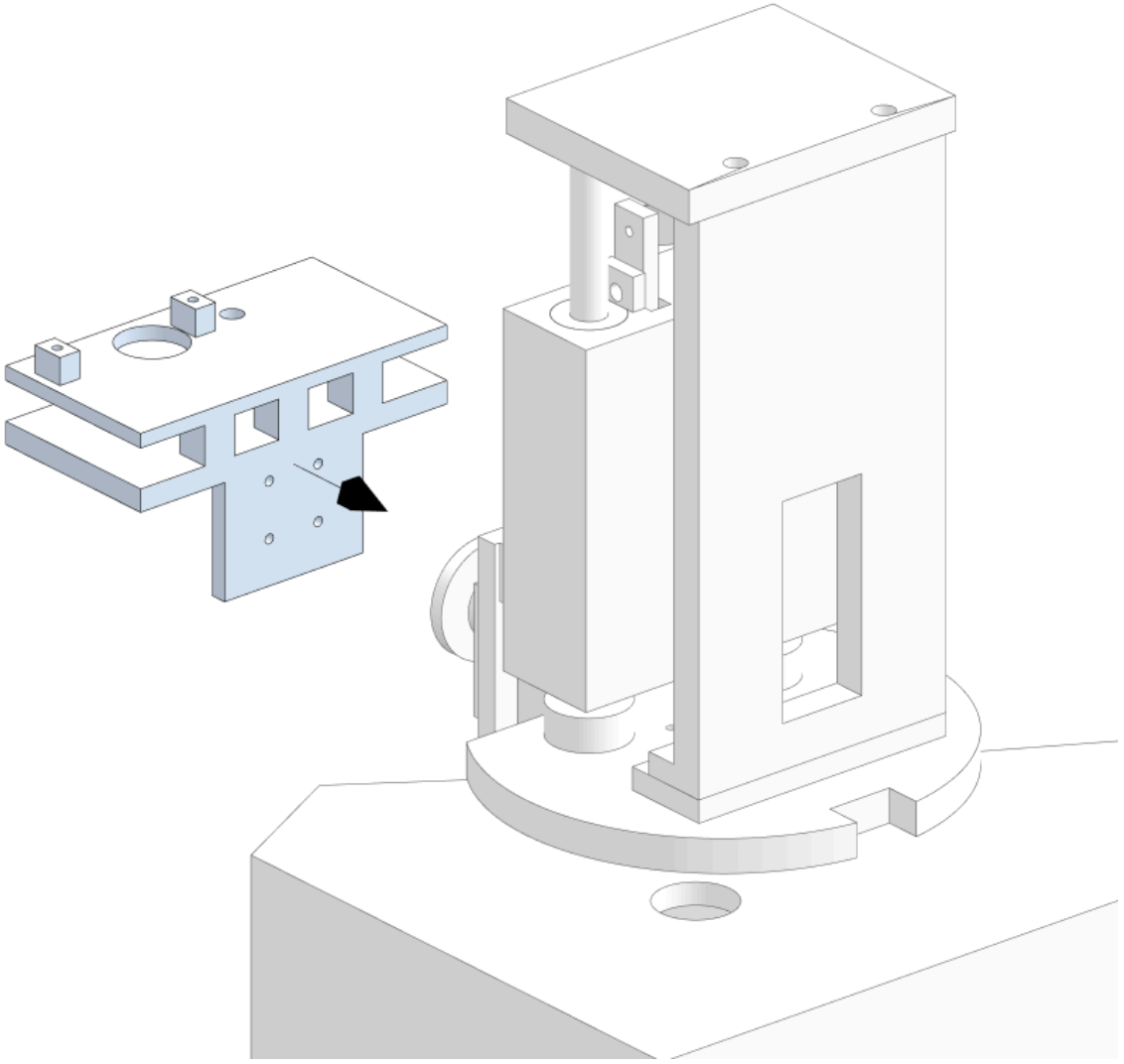
`w 0`



16

Gripper frame

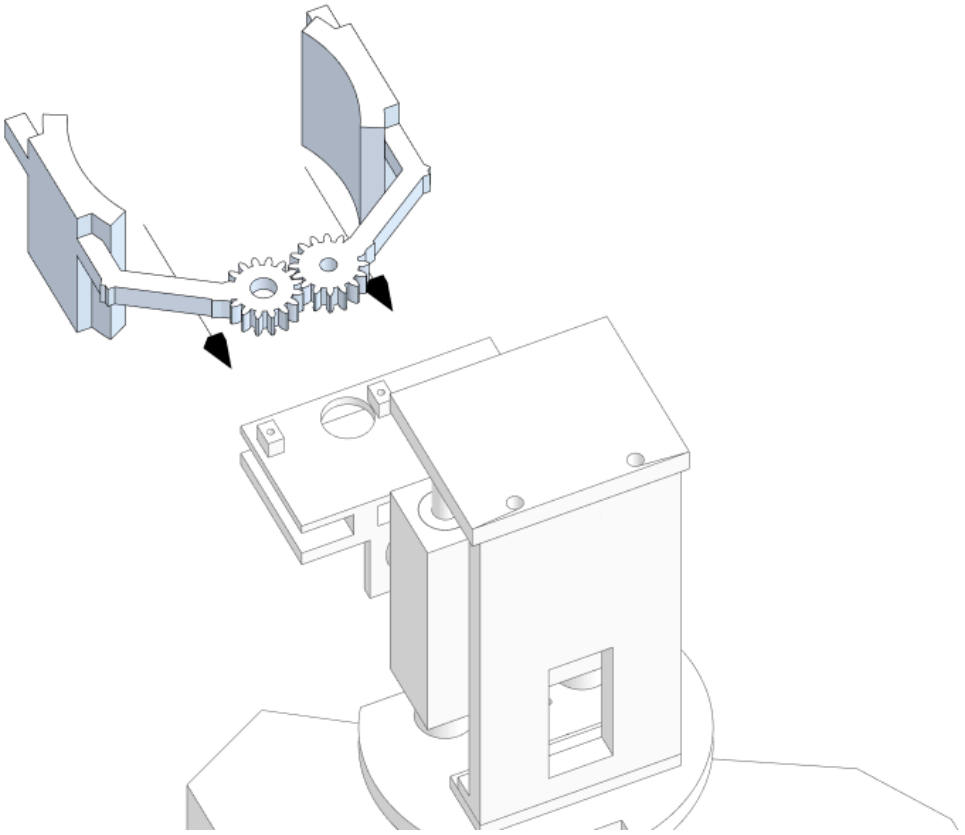
4× servo-bag



Upright: plates level, opening facing out

17

Both fingers, gears meshed

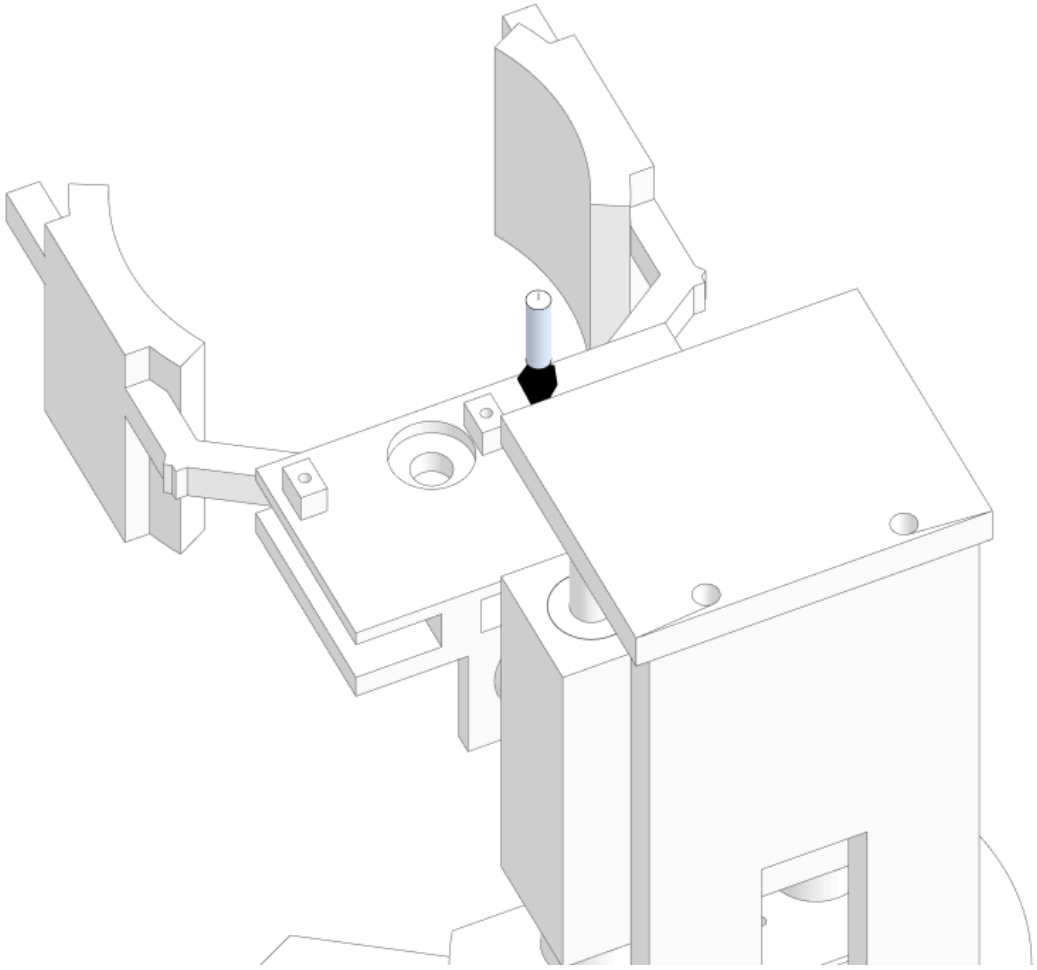


Slide them in together between the plates, open by the same angle

18

Right finger axle

1× #6-32 × ½″



! From the top through the right finger; it must swing freely

19

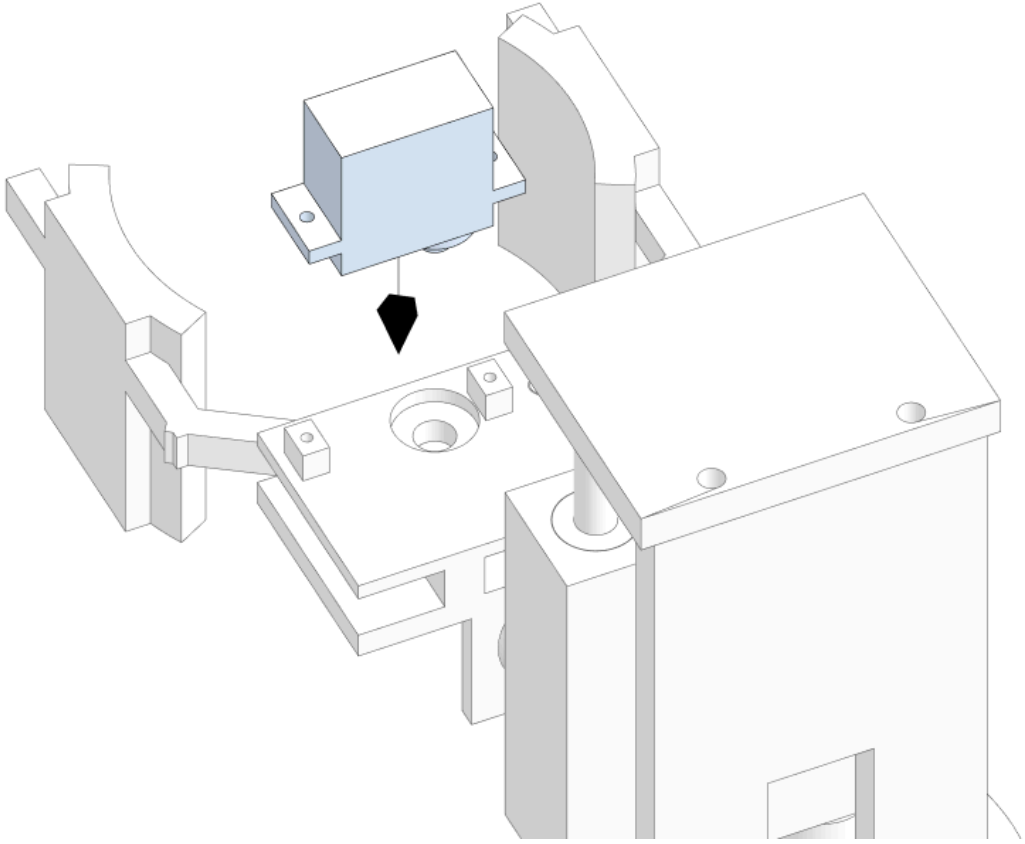
Gripper servo

3×  servo-bag

SG90

20°

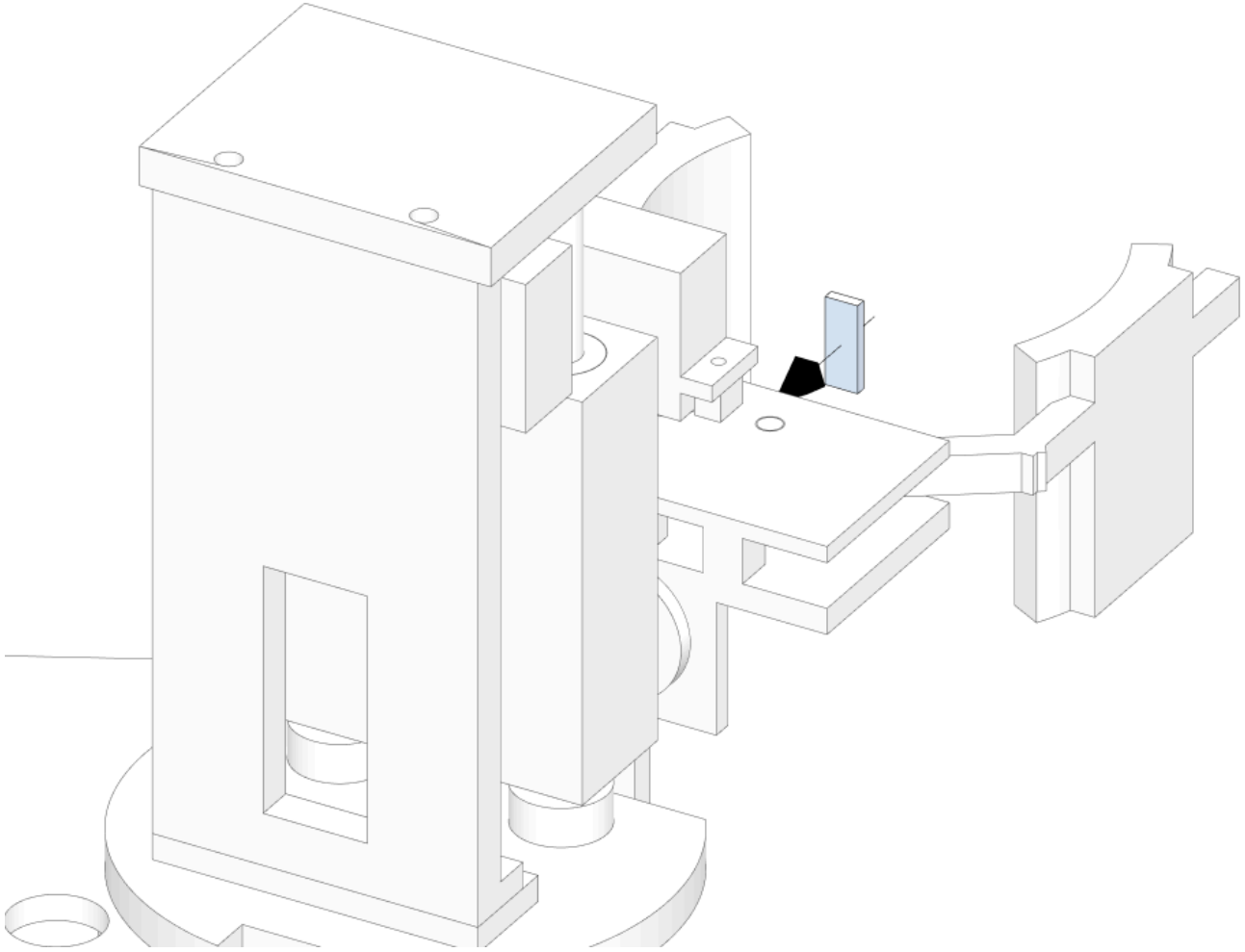
g 20



Spline down into the left finger's socket (it is the horn)

20

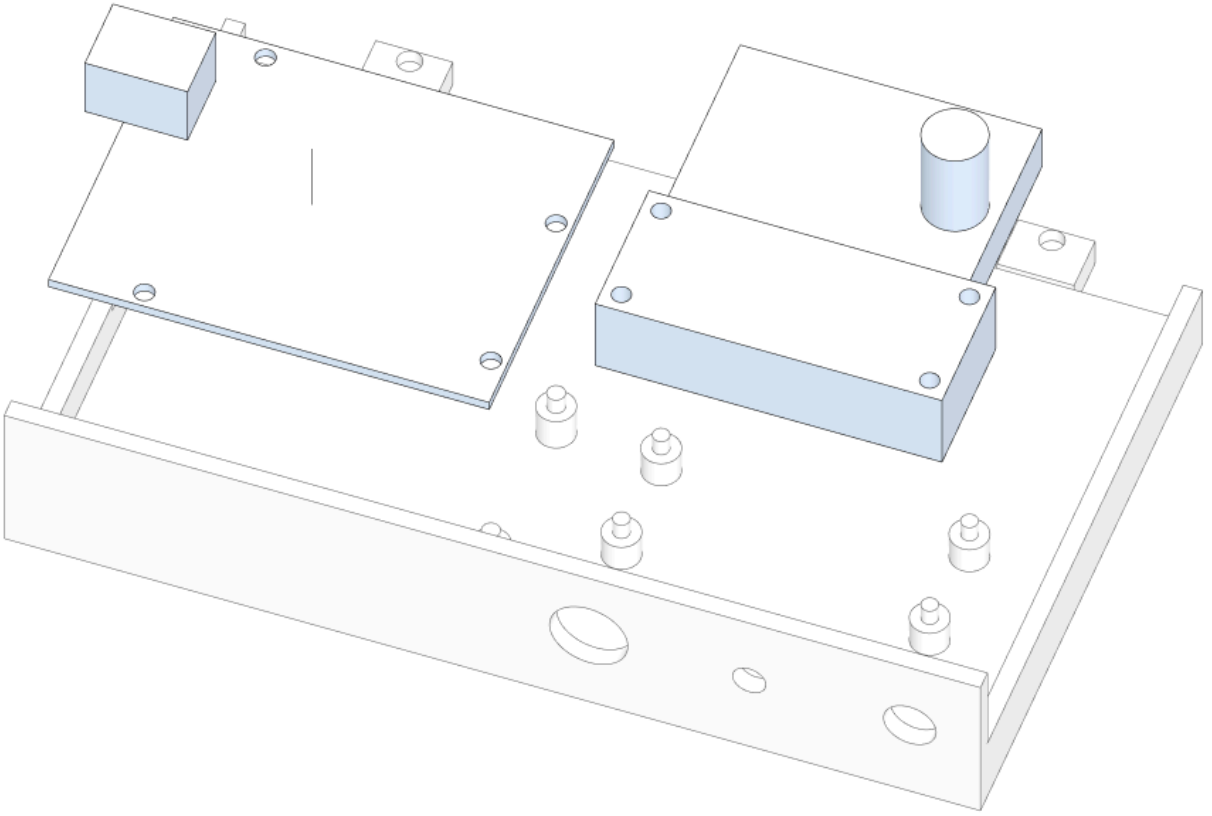
Can bumper



Silicone or rubber, 9 × 13.5 mm, superglue to the frame

21

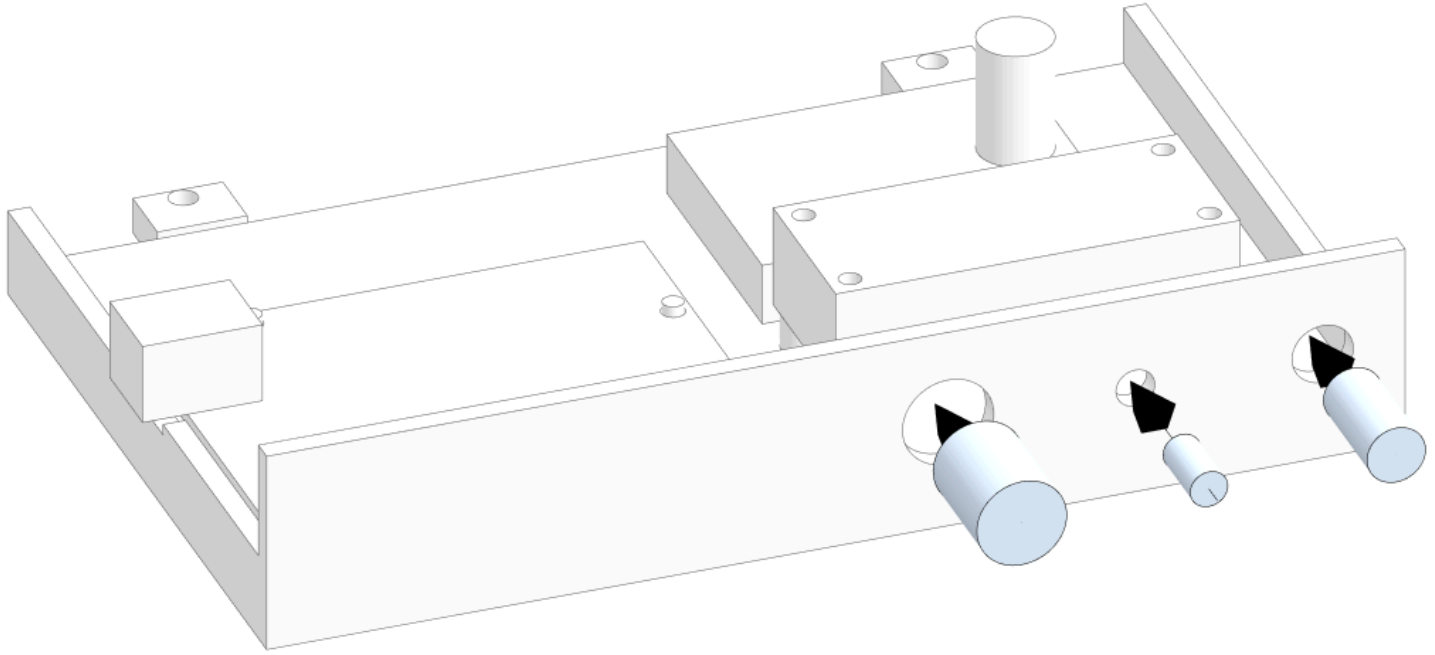
Electronics on the tray



! Pegs + a dab of hot glue; breadboard is self-adhesive. Buck already set to 5.8 V

22

Panel: button, LED, power

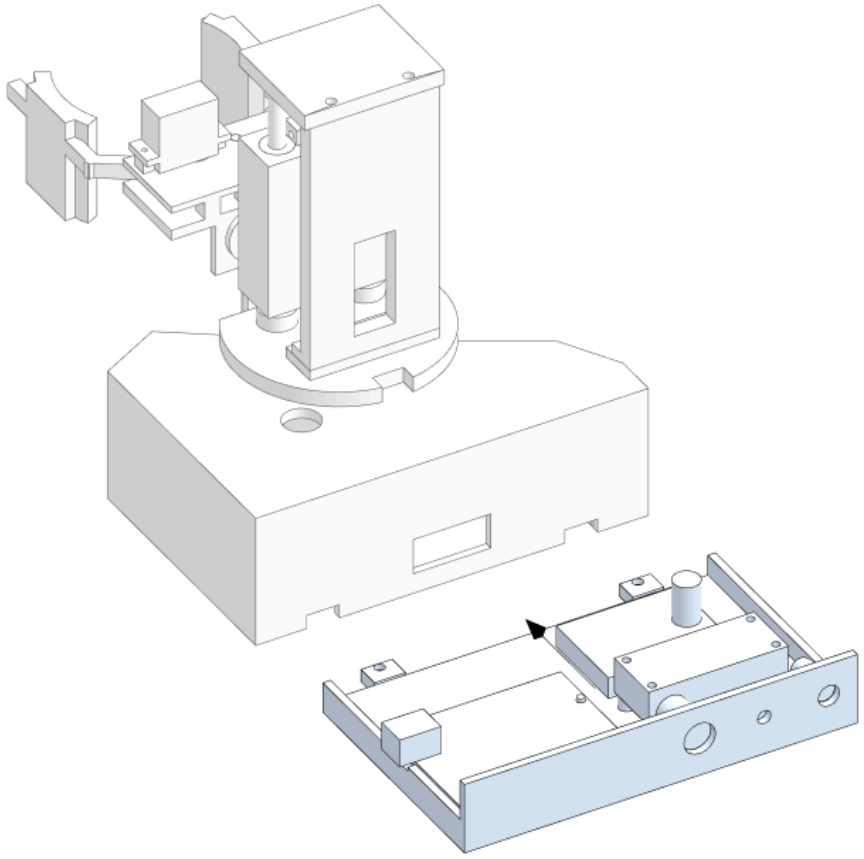


Through the panel holes from the front, nuts inside. Power: DC jack, or the LIZVOR terminal-block tip

23

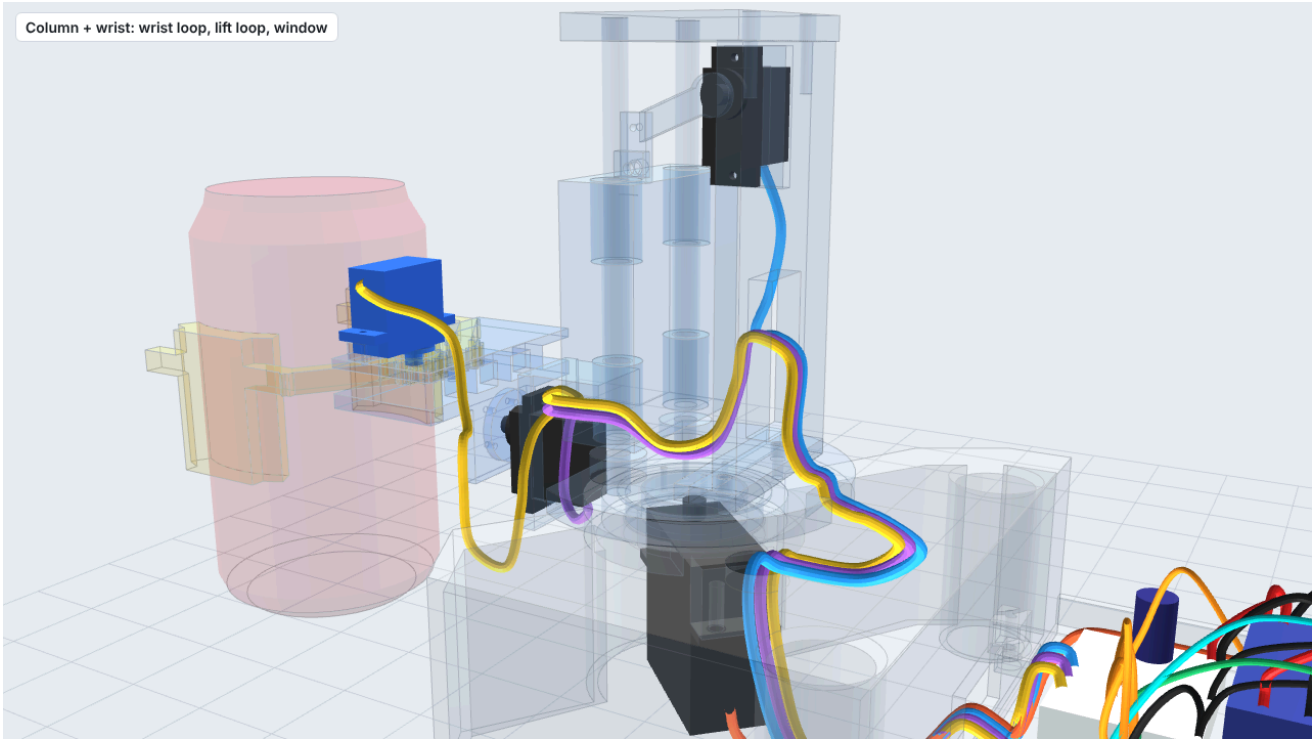
Tray onto the base

2× #6-32 × ½″



Tabs slide under the base wall, screws from below

24 Wiring



Servo lead	Pin
MG996R turntable	D9 (+/− on the buck)
MG90S lift	D10
MG90S wrist	D6
SG90 gripper	D5
START button	D4 + GND
LED + 220 Ω	D13 + GND

Route: claw → wrist loop → along the arm → lift loop → spine window → down the back → turret notch → turn loop → base hole → tray. **Tie at the tie points, leave the 3 loops free.** Wrist loop ≥ 70 mm of cable.

3D view: viewer/wiring_360.html · moves: viewer/wiring_motion.html

(!) Power off while wiring. Servo plugs: brown = GND, red = +5.8 V, orange = signal.

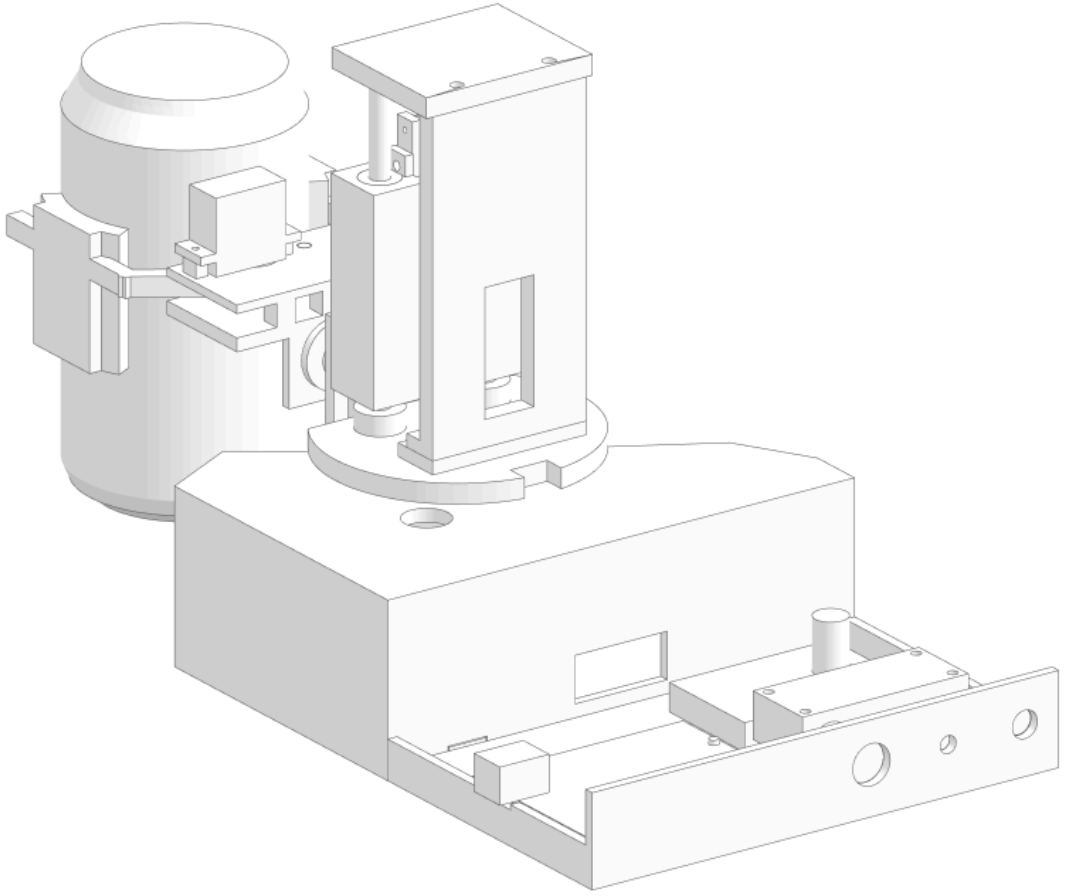
25

Test

1. Upload

`firmware/concept_a`

2. Can on START, 5 in out
3. Press START
4. Can lands upside down at TARGET



If a move is off: re-set its angle with servo_setup, then edit the constant at the top of concept_a.ino.